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2021

Spreading System 3.0

Repair Training

Contents

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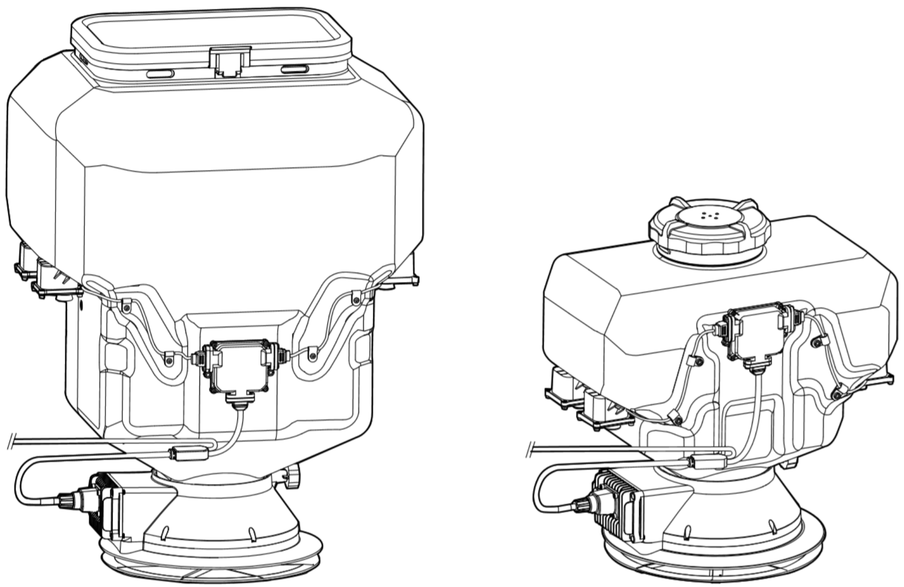
01 Product Introduction

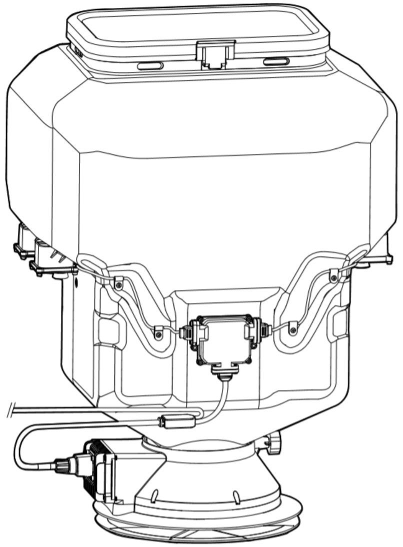
01 Product Introduction — T30/T10 Spreading System 3.0



Product Name: T30 & T10 Spreading System 3.0

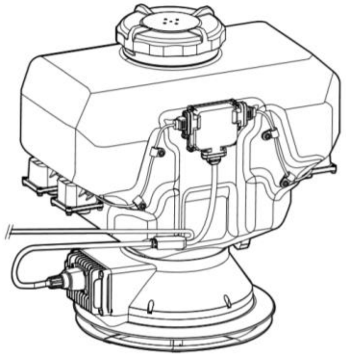
- The T30/T10 Spreading System 3.0 is compatible for mounting with the Agras T30 and T10 drones, offering efficient, reliable, and stable spreading operations.
- The Spreading System 3.0 has a built-in stirring device, which can prevent the material blockages and improve operating accuracy and reliability.
- With the weight sensor and control module, the spreading system can detect and monitor the remaining materials in the tank in real-time and control the material delivery rate, making empty tank warnings more accurate.

01 Product Introduction — Product Comparison — Spreading System



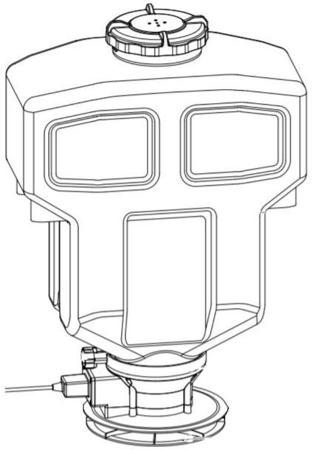
T30 Spreading System 3.0

- Spread materials 360° around the aircraft
- Real-time monitoring of the remaining materials in the tank
- SN: 4EP



T10 Spreading System 3.0

- Spread materials 360° around the aircraft
- Real-time monitoring of the remaining materials in the tank
- SN: 4EN



T Spreading System 2.0

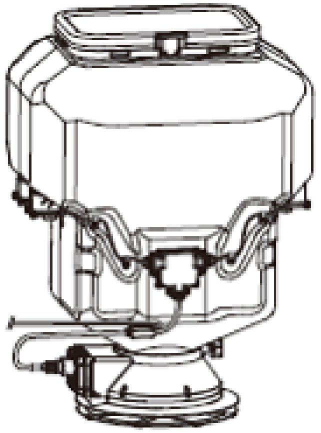
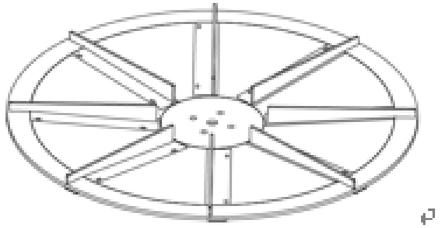
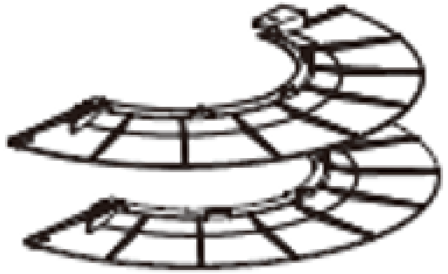
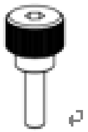
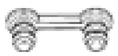
- Spread materials 360° around the aircraft
- Empty tank warning
- SN: 3E8

01 Product Introduction — Specifications

Product	T30 Spreading System 3.0	T10 Spreading System 3.0	T Spreading System 2.0
Compatible Aircraft	Agras T30	Agras T10	Agras T16/T20
Spreader Weight	4.1 kg	3.0 kg	2.1 kg
Compatible Material Diameter	0.5-5mm	0.5-5mm	0.5-5mm
Spread Tank Volume	40L	12L	20L
Spread Tank Internal Load	40kg	10kg	Using with the T16: 16 kg Using with the T20: 20 kg
Spreading Range*	5 - 7 m	5 - 7 m	5 - 7 m

*Varies according to material diameter, spinner disk rotating speed, hopper outlet size, and flight altitude. For best operating performance, it is recommended to adjust the corresponding variables to achieve a spreading range of 5-7 meters.

01 Product Introduction — In the Box

Spreading System 3.0 × 1	Spare Spinner Disk × 2	Fender Pair × 1
		
Fender Screws × 4	Stopper	
	 (for T30 only)  (for T10 only)	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02 Material Identification

02 Material Identification — Main Module

Control Module

Core module of the spreading system.

Servo Module

Controls the change of the hopper gate angle and the opening/closing of the hopper gate.

Reduction Gearbox

Drives the stirring device to spin, brings the materials to the hopper gate, and spreads them out of the hopper gate.

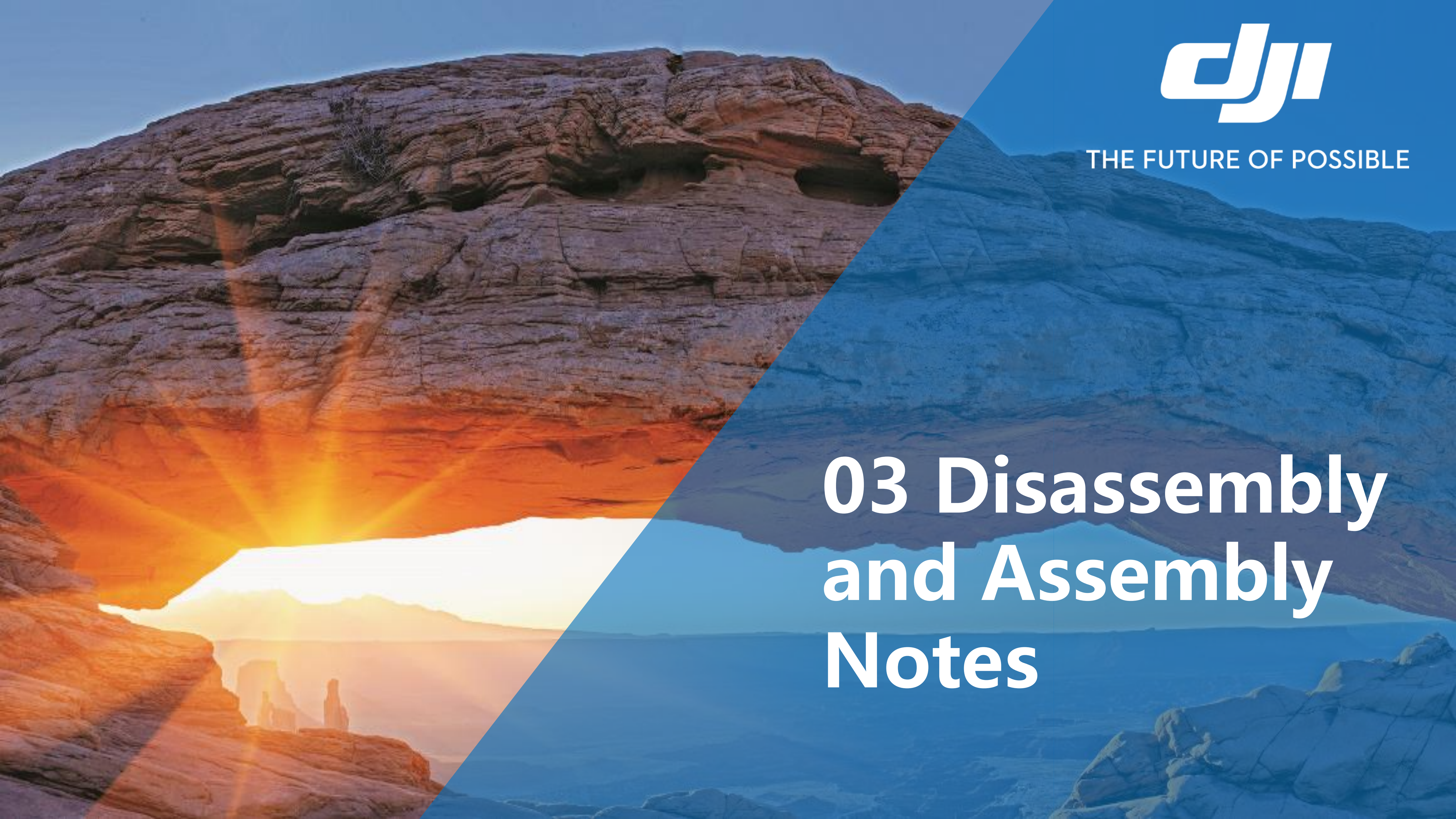
02 Material Identification — Main Module

Weight Sensor

Responsible for collecting the weight of the tank.

Weight Collection Board

Receives the data from the weight sensor and calculates the weight.



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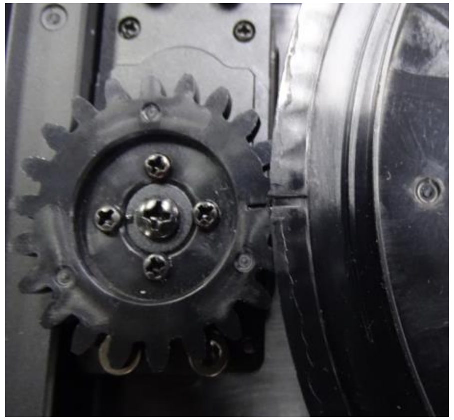
03 Disassembly and Assembly Notes

03 Disassembly and Assembly Notes



When installing the large gear, make sure that the large gear is aligned with the notch of the small gear on the servo module.

On the weight control module, the number of the weight sensor modules on the left side is 1 and 2, and that of weight sensor modules on the right side is 3 and 4. (The first three digits of the serial number for No.1 is 4JT, the first three digits of the serial number for No.2 is 4QG, the first three digits of the serial number for No.3 is 4QH, and the first three digits of the serial number for No.4 is 4QJ.)

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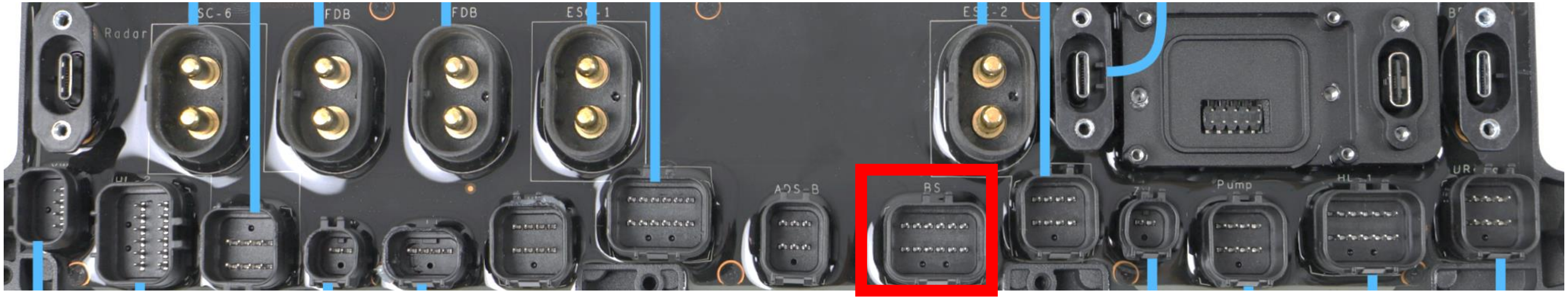
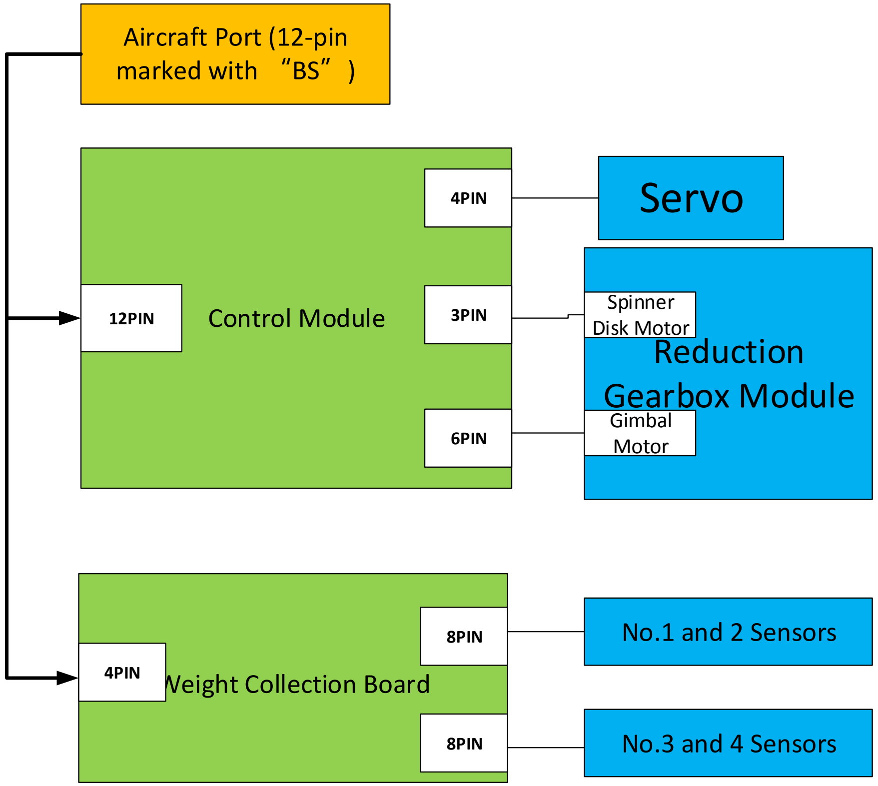
05 Principles Diagram and Boards Introduction

05 Principles Diagram





Chip	Function
Control Module	Core module of the spreader; collects Hall sensor signal, ESC signals, and servo signals, and controls the various resources of the entire spreader.
Weight Collection Board	Receives the data from the weight sensor and calculate the weight, then transmits the data to the control module for app prompts of an empty tank.
Weight Sensor	Collects the weight of the tank.
Servo	Controls the change of the hopper gate angle and the opening/closing of the hopper gate.
Reduction Gearbox Module	There are two built-in motors. The gimbal motor drives the structure attached above the motor to rotate. The control module determine whether to make an app prompt for an empty tank based on the current value of the motor; the spinner disk motor drives the spinner to rotate and spreads the materials that fall on the spinner disk.

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06 Test Plan & Fixtures

06 Test Plan & Fixtures (Spreader)

No.	Module	Test	Content	Device/ Fixture	Testing Environment	Test Software (Name/Version)	Damage Assessment	Repair
1	Spreading System	Firmware Update	Update the firmware.	/	Outside of SWS	Official Assistant	No	Yes
2		App Functionality Test	Check the app functions.	/	Outside of SWS	App	Yes	Yes
3		Writing the Weight Value K	Write the value K.	/	Outside of SWS	App	Yes	Yes

The value K needs to be written again when the load sensor is replaced.

06 Test Plan & Fixtures (Tests after components replacement)

Product	Module	Test Items	Tests Required After Component Replacement					
			Spreader					
			Control Module	Load Collection Board	Load Sensor	Reduction Gearbox	Servo	Cable
Spreader	Whole Product	Writing the Weight Value K	x	√	√	x	x	x
		Firmware Update	√	√	√	√	√	√
		App Functionality Test	√	√	√	√	√	√

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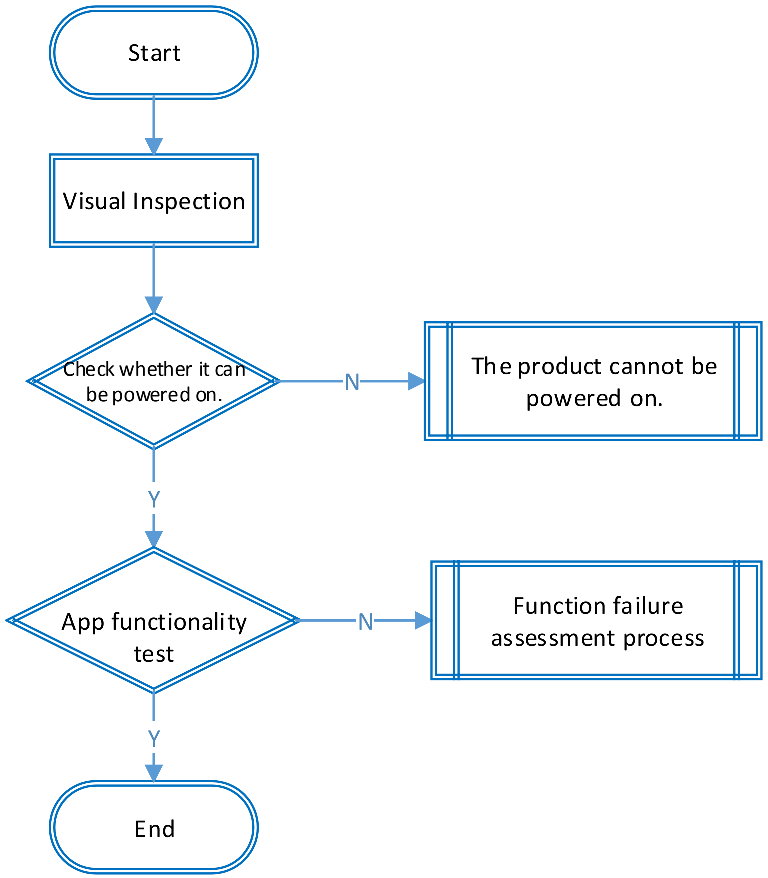
07 Malfunction Assessment and Analysis

07 Malfunction Assessment and Analysis — Common Issues

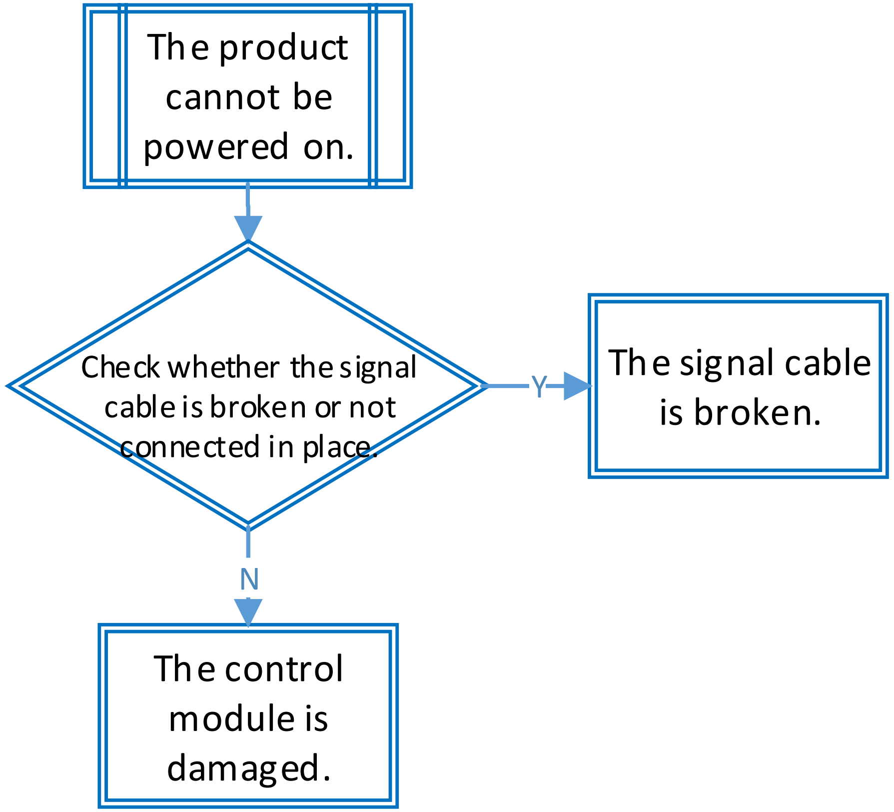
Check whether there is any cosmetic damage on the product, whether the product has crashed, whether any boards are water damaged, and whether all the components are intact.



07 Malfunction Assessment and Analysis — Common Issues

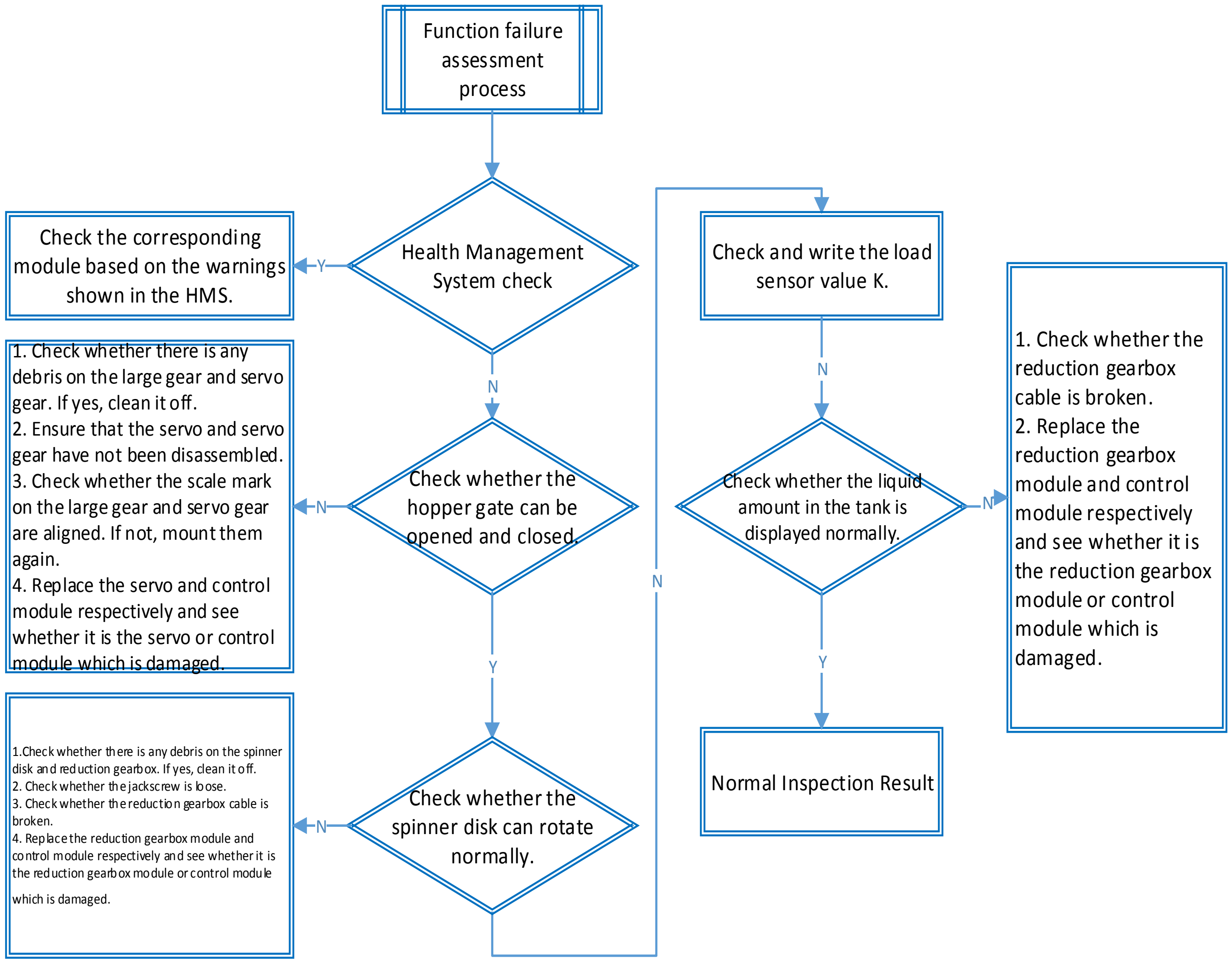
If the product cannot be powered on, check the following modules:

1) Control module
2) Spreader signal cable



07 Malfunction Assessment and Analysis — Common Issues

Link the aircraft with the T30 remote controller, and use the app to check whether the hardware connection is normal. If there are any warnings shown in the app or the module cannot be turned on because the module cannot be identified, determine that the corresponding module is damaged.

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FAQ

Compatibility List

Motor

Material Name	Material Number	T16	T20	T10	T30
10015 Motor (Thickness)	BC.AG.AA000137.02	√	×	×	×
10015 Motor (Thin)	BC.PR.AA000090.01	×	√	×	×
10010S Motor	BC.AG.AA000336.01	×	×	√	×
10018 Motor	BC.AG.AA000335.01	×	×	×	√
Silicone Rubber Pad	YC.XJ.QT000196.04	√	√	√	√
Teflon Gasket	YC.SJ.WS002416.03	√	√	√	√
Aluminum Alloy Propeller Holder CCW	YC.JG.QX000240.01	×	√	√	√
Aluminum Alloy Propeller Holder CW	YC.JG.QX000238.03	×	√	√	√
33-inch Propeller CCW	YC.SJ.GG000060.01	√	√	√	×
33-inch Propeller CW	YC.SJ.GG000061.01	√	√	√	×
38-inch Propeller CCW	YC.JG.ZS000813.01	×	×	×	√
38-inch Propeller CW	YC.JG.ZS000812.01	×	×	×	√

Aircraft

Module	T16	T20	T10	T30
Aerial-Electronics System Module	x	x	√	√
Spraying Module	x	x	√	√
RF Module	x	x	√	√
RTK Module	x	x	√	√
Power Distribution Board	x	x	x	x
Cable Distribution Board	None	None	x	x
Radar	x	x	√	√
Delivery Pump	√	√	x	x
Electromagnetic Exhaust Valves	None	x	x	x
FPV Module	x	x	√	√
Liquid Level Gauge	x	x	x	x
Upward Radar	None	None	x	x
ESC	x	x	√	√
Remote Controller	x	x	√	√

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Thank You